



Republic of the Philippines
Department of Agriculture
BUREAU OF FISHERIES AND AQUATIC RESOURCES
REGIONAL FISHERIES LABORATORY XII
J. Catolico St., Lagao, General Santos City

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Document Code: SOP-09-02	INTERNAL QUALITY CONTROL FOR MICROBIOLOGICAL TESTS		

This document provides guidelines in implementing a comprehensive internal quality control program to ensure the reliability of results in microbiological tests.

1. Responsibilities

1. Technical Manager
2. Quality Assurance Manager
3. Laboratory Analyst

2. Definitions

Culture Media. Formulations of substances, in liquid, semi-solid, or in solid form, which contain natural and/or synthetic constituents intended to support the multiplication, or to preserve the viability of microorganisms. (*Note Introduction*

: *This includes diluents and other suspending fluids.*)

Quality Control. The final inspection and testing of the finished product to ensure its compliance with predetermined performance criteria.

Lot of Culture Media. Fully traceable unit of a raw material referring to a defined amount which is consistent in type and quality and having been assigned the same lot number.

Batch of Culture Media. Fully traceable unit of a medium referring to a defined amount of semi-finished or end-product, which is consistent in type and quality and which has passed the requirements of production and quality assurance testing, and which has been produced within one defined period, having been assigned the same batch number.

Performance of Culture Media. The response of a culture media to challenge by test organisms under defined conditions.

Test Organisms. These are microorganisms generally used for quality control and performance testing of culture media.

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Reference Strain/Culture. A microorganism defined to at least genus and species level, catalogued and described according to its characteristics and stating its origin.

Reference Stocks (Master Culture). A set of separate cultures obtained in the laboratory by a single subculture from the reference strain.

Working Cultures. A primary sub-culture from a reference stock.

Sub-cultures (Passage). The transfer of established microorganism from viable culture to fresh medium with growth of the microorganism.

3. Maintenance of Cultures used for Quality Control Testing

3.1 Reference Strains

- 3.1.1 Obtain Passage 0 or 1 reference cultures from reliable sources such as ATCC, Microbiologics, Philippine National Collection for Microorganism (PNCM Biotech) or other commercial sources.
- 3.1.2 After receipt of reference culture in the laboratory, activate the culture by growing it in appropriate liquid medium. Incubate at optimum temperature of growth for 18 – 24 hours.
- 3.1.3 Test for purity.
- 3.1.4 Verify the identity of the reference culture.
- 3.1.5 Record results of test for purity and identification in LF W09-02-01 (Maintenance of Cultures).
- 3.1.6 If pure, use the growth from the liquid medium to prepare the culture in frozen glycerol and broth with beads (Cryobank).
- 3.1.7 Label and store the culture at -18°C.
- 3.1.8 The preserved culture generated by this process is termed **MASTER CULTURE** and should not be accessed frequently. This can be stored for years as long as it is still viable.

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3.2 Stock Culture

- 3.2.1. Aseptically transfer 1 bead from the master culture into appropriate liquid medium and incubate at optimum temperature of growth for 18 – 24 hours.
- 3.2.2. Pipet 1 ml of the broth with growth into Cryovials with glycerol and beads. Prepare at least three to four vials to last for 3 – 12 months.
- 3.2.3. Label and store the culture at -18°C. This can be stored for 0.5 to 2 years.
- 3.2.4. The preserved culture generated by this process is termed **STOCK CULTURE** and may be accessed to prepare WORKING cultures.

3.3 Working Culture

- 3.3.1 Aseptically transfer 1 bead from the stock culture into appropriate non-selective medium and incubate at optimum temperature of growth for 18 – 24 hours.
- 3.3.2 Label and store at 0-4°C up to 4 weeks. Discard after this time.
- 3.3.3 Use culture for media growth performance checks or test method controls.

Notes:

- Reference culture may only be sub-cultured once. Growing a reference culture from its preserved status (frozen or lyophilized) is not a subculture/passage.
- Stock culture may be sub-cultured provided that the preceding working culture must not exceed 5th passage.
- Working culture must not be sub-cultured to replace the reference stock.
- The subsequent growth on the fresh media constitutes another subculture or passage.

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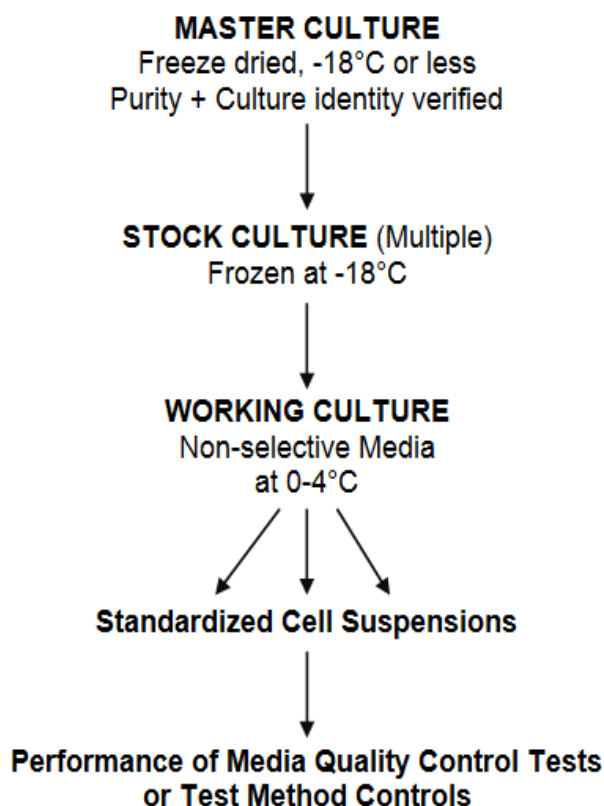


Figure 1. Maintenance of Cultures

Quality Control for Microbiological Tests

- 4.1 Use of negative and positive controls during testing is necessary to ensure that any growth in the media comes from the sample itself.
- 4.2 Negative Controls. The culture medium used is spiked with a working culture of test organism which is expected to have a negative growth and incubated at an appropriate time and temperature together with the samples. Negative control organisms used for each test method are listed below:

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- Aerobic Plate Count / Heterotrophic Plate Count – No negative control
- Total Coliform, Fecal Coliform and E. coli Count – *Staphylococcus aureus*
- *S. aureus* Count – *Escherichia coli*
- *Salmonella sp.* Detection - *Staphylococcus aureus*
- *Shigella sp.* Detection - *Staphylococcus aureus*

4.3 Positive Controls. The culture medium used is spiked with a working culture of test organism which is expected to have a positive growth/result and incubated at an appropriate time and temperature together with the samples. Positive control organisms used for each test method are listed below:

- Aerobic Plate Count / Heterotrophic Plate Count – *Escherichia coli*
- Total Coliform, Fecal Coliform and E. coli Count – *Escherichia coli*
- *S. aureus* Count – *Staphylococcus aureus*
- *Salmonella sp.* Detection – *Salmonella typhimurium*
- *Shigella sp.* Detection – *Shigella sonnei*

4.4 QC results for negative and positive controls are recorded in LF W04-MIC-02 (Analyst Worksheet for the Bacteriological Examination of Water) and LF W04-MIC-03 (Analyst Worksheet for the Bacteriological Examination of Fish and Fishery Products).

5. Quality Control of Microbiological Media

Microbiological media includes dehydrated culture media, laboratory's freshly prepared media, reagents and reagent water. Quality control is conducted to verify satisfactory level of freedom from contamination, demonstrate the appropriate performance of the medium and assurance against physical imperfections (e.g. color, pH, temperature) that may compromise the utilization of the media.

5.1 Management and Storage of Culture Media

5.1.1 Purchase of culture media is planned to encourage proper turnover of stocks (first-expiry-first-out).

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- 5.1.2 Upon receipt of culture media in the laboratory, check for seal and record expiry dates, manufacturer's lot/batch numbers, quantity received and initial of person receiving and placing the items in stock in LF W09-02-02 (Lists of Received Media, Reagents and Consumables).
- 5.1.3 Store media in a clean, dry cabinet. Temperature of the room should be 20 – 25°C. For other media that need to be refrigerated, storage temperature should be based per manufacturer's instruction.
- 5.1.4 Perform quantitative and/or qualitative tests specific for every representative type of media received and record results in LF W09-02-03 (Quality Control of Culture Media).

5.2 Tests for the Quality of Reagent Water

- 5.2.1 The quality of reagent water used for microbiological tests and media preparation is monitored by conducting HPC and pH check.
- 5.2.2 Heterotrophic Plate Count (HPC) of reagent water is conducted monthly. Acceptable result is <500 cfu/ml.
- 5.2.3 The pH of reagent water is checked weekly. Acceptable pH should be within 5.5 – 7.5.
- 5.2.4 Results are recorded in LF W09-02-04 (Reagent Water Quality Monitoring).

5.3 Preparation of Culture Media and Reagents

- 5.2.5 Use clean separate spatula for weighing each type of media.
- 5.2.6 Prepare medium in container at least two times its volume.
- 5.2.7 Use clean and pre-wet graduated cylinder or pipettes in measuring volumes.
- 5.2.8 Record the type of media prepared, unique identity (batch no.), manufacturer's identity and lot/batch number, date of preparation and identity of person who prepared, weight and volume of media/solutions made and pre and post-sterilization pH and physical appearance in LF W04-MIC-01 (Reagent and Media Preparation Logbook).

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- 5.2.9 Sterilization temperature, time and pressure for each batch of media prepared are recorded in LF W04-MIC-04 (Sterility Check).
- 5.2.10 Volumes dispensed and post-sterilization volume check is made for media which are used as diluent or the volume is critical for other reasons. Details are recorded in LF W09-02-05 (Media Volume Check).

5.3 Test Procedures for Media Quality Control

- 5.3.1 Suspend 3-5 isolated colonies in a small volume of suitable medium and use growth from an 18-24 hour culture of the quality control organism. Adjust the turbidity to approximate a McFarland 0.5 turbidity standard.
- 5.3.2 For testing the nutritive capacity of a medium, inoculate each test plate with a calibrated or disposable loop loaded with diluted suspension to provide $10^2 - 10^3$ cfu/plate.
- 5.3.3 For testing the inhibitory capacity of a selective medium inoculate each test plate with a calibrated or disposable loop to provide $10^4 - 10^5$ cfu/plate.
- 5.3.4 For testing the performance of liquid medium for its nutritive capacity a cell suspension should be prepared so that the chosen aliquot will deliver approximately $10^1 - 10^2$ cfu per unit of test medium.
- 5.3.5 For testing the performance of liquid medium for its inhibitory capacity, heavier Inocula of the order of $10^4 - 10^5$ cfu will normally be used.
- 5.3.6 Incubate the inoculated test media under conditions specified in the relevant test method.

5.4 Parameters to be Measured in Media Quality Control

5.4.1 Productivity

Where it is necessary to demonstrate the growth of a microorganism in a medium, the productivity should be measured.

- For **quantitative** methods the Productivity Ratio P_R is determined as follows:

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$P_R = N_S / N_0$ where:

N_S is the total colony count obtained on the tested culture medium

N_0 is the total colony count obtained on the defined reference culture medium. It should be ≥ 100 cfu.

- For **qualitative** evaluations, visual checks are carried out and growth scores allocated (e.g. “0” corresponds to no growth, “1” corresponds to weak growth, “2” corresponds to good growth).

5.4.2 Selectivity

Where it is necessary to demonstrate that a medium suppresses the growth of a microorganism, the selectivity should be measured.

- For **quantitative** methods, the Selectivity Factor S_F is calculated quantitatively as follows:

$$S_F = D_0 - D_S \quad (S_F, D_0 \text{ and } D_S \text{ are expressed in log}_{10} \text{ units})$$

D_0 is the highest dilution showing growth of at least 10 colonies on the non-selective reference medium

D_S is the highest dilution showing comparable growth on the test medium

Note: The S_F of non-target microorganisms on most selective media should be at least 2.

- For **qualitative** methods the unwanted strain(s) should be inhibited partly or completely.

5.4.3 Specificity

The specificity is given by essential characteristics that differentiate related organisms – by the presence, absence and/or grade of expression of biochemical responses and colony sizes morphology.

5.5 Growth Recovery of Control Microorganisms

For testing commercially available dehydrated media and finished media (for enumeration), quantitative testing is recommended. Qualitative testing is sufficient for other types of media.

5.5.1 Quantitative Recovery

5.5.1.1 Non-selective media

- Perform viable counts on both test and reference medium;

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- Compare the results as described in 5.5.1. The P_R should be calculated using the counts from both media. An acceptance criteria of at least 70% recovery is recommended.
- The medium also needs to be assessed for typical morphology to complete the performance evaluation on the medium.

5.5.1.2 Selective solid media

- Perform viable counts on both the test and reference medium;
- Compare the results as described in 5.5.1. The P_R should be calculated using the counts from both media. An acceptance criteria of at least 50% recovery is recommended.
- The medium also needs to be assessed for typical morphology to complete the performance evaluation on the medium.
- It is also relevant to demonstrate the capacity of the test medium to suppress the negative control organism. The S_F of negative control microorganisms on most selective media should be at least 2.

5.5.1.3 Non-selective liquid media

- Between 10^1 - 10^2 cfu of the test organism is inoculated into the test broth, incubated and then a standard aliquot is removed to enumerate by quantitative methods, to demonstrate the recovery of an adequate number of test organisms.

5.5.1.4 Selective liquid media

- Test organism: 10^1 - 10^2 cfu is inoculated into the test broth, reference broth (TSB or BHI) and solid reference medium. The solid reference medium is use to confirm the cfu in the inoculum.
- Negative control organism: 10^4 - 10^5 cfu is inoculated into a second set of the same media.
- Test and negative control organism as a mixed culture: 10^1 - 10^2 cfu of the test organism and 10^4 - 10^5 cfu of the negative control organism is inoculated into a third set of media. The solid reference medium here should be a non-

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selective agar that allows differentiation of test organisms and negative control organism.

- After incubation, remove a measured volume from each broth and spread on solid non-selective media.
- After incubation of solid media, determine the percentage recovery for the test organism and the degree of inhibition for the negative control organism. For the mixed culture, the % recovery of the positive organism should not be compromised.

Selective Liquid Medium Testing

Test Medium			Non Selective Reference Medium		
+ve control org. 10 ¹ -10 ² cfu ↓ subculture Non-Inhibitory Medium (±Indicator) Count	-ve control org. 10 ⁴ -10 ⁵ ↓ subculture Non-Inhibitory Medium (±Indicator) Count	Mix(+ve &-ve) ↓ subculture Non-Inhibitory Medium (±Indicator) Count Both Org.	+ve control org. 10 ¹ -10 ² cfu ↓ subculture Non-Inhibitory Medium (±Indicator) Count	-ve control org. 10 ⁴ -10 ⁵ ↓ subculture Non-Inhibitory Medium (±Indicator) Count	Mix(+ve &-ve) ↓ subculture Non-Inhibitory Medium (±Indicator) Count Both Org.
Determine: % Recovery of the +ve organism % Inhibition of -ve organism % Recovery of the +ve organism from mix should equal or exceed from +ve organism alone.					

5.5.2 Qualitative Recovery

An ecometric method (Figure 1) is used to distribute CFUs over the plate. The growth (number of streak lines or quadrants grown) for both test and reference media is compared and the growth index G_i calculated or determined. If a 21 streak line plate is prepared, then the number of streak lines on the reference medium is recorded as the Absolute Growth Index (AGI), while the number of streak lines on the test medium is recorded as the Relative Growth Index (RGI).

The % Relative Growth Index is calculated as follows:

$$\% \text{ RGI} = \text{RGI} / \text{AGI} \times 100$$

5.5.2.1 Non-selective solid media

- Perform viable counts on both test and reference medium.

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- Compare the results as described in 5.6.2. Calculate the % RGI using the counts from both media. An acceptance criteria of a % RGI of at least 70% is recommended.
- The medium also needs to be assessed for typical morphology to complete the performance evaluation on the medium.

5.5.2.2 Selective solid media

Perform viable counts on both test and reference medium.

- Compare the results as described in 5.6.2. Calculate the % RGI using the counts from both media. An acceptance criteria of a % RGI of at least 50% is recommended for the test organism.
- The medium also needs to be assessed for typical morphology to complete the performance evaluation on the medium.
- Demonstrate the capacity of the test medium to suppress the negative control organism. The recommended acceptance criteria for negative control microorganisms on most selective media is <25%.

5.5.2.3 Non-selective liquid media

- Between 10^1 - 10^2 cfu of the test organism is inoculated into the test broth, incubated and then a standard aliquot is removed to enumerate by quantitative methods, to demonstrate the recovery of an adequate number of test organisms.
- Use standard inocula of working cultures and inoculate directly into the medium being tested and a reference broth. Qualitative evaluation is carried out visually by allocating growth scores as follows: zero turbidity or 0, very light turbidity or 1, good turbidity or 2. Score for the wanted microorganisms should be good turbidity or 2. Other characteristics such as gas formation, color change, etc. should also be assessed.

5.5.2.4 Selective liquid media

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- Inoculate a test broth with positive control bacterium, another test broth with negative control bacterium, and a third test broth with a mixture of positive and negative control bacteria.
- After incubation, a standard loop (10 ul) from the test broths for the positive bacterium and the mixture are plated out onto a selective medium for growth of the positive bacterium.
- A standard loop (10 ul) from the test broth for the negative control bacterium is plated onto a non-selective medium.
- The test medium is considered to have passed if at least 10 colonies of the positive control develop on the selective medium and no growth or less than 10 colonies of the negative control develop on the non-selective medium.

5.6 Recording and Interpretation of Results

- 5.6.1 Record all results of media quality control in LF W09-02-03 (Quality Control of Culture Media) and interpret results according to the recommended guidelines.

A medium's performance is regarded as satisfactory if all test strains grow or are inhibited as appropriate from the medium being tested and colonial morphology and reactions produced in the medium are typical for the organism on that particular type of medium.

The laboratory's media specifications for acceptance and rejection criteria should be documented to be able to accept all batches of "satisfactory" laboratory produced medium.

In cases wherein there is rejected batch of dehydrated medium, the laboratory should discard it immediately.

5.6.2 Recommended Results for Quantitative Recovery

Productivity: $\geq 70\%$ (wanted organism) nonselective medium
 $\geq 50\%$ (wanted organism) selective medium
 $\leq 25\%$ (unwanted organism)

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Selectivity: >2 (Log)
Specificity: Reject if fails to produce typical colonial morphology, size or biochemical response
Reject if fails to suppress background flora

5.6.3 Recommended Results for Qualitative Recovery

% RGI: ≥70% (wanted organism) nonselective medium
≥50% (wanted organism) selective medium
≤25% (unwanted organism)

Growth Observed: good growth (wanted organism) or zero/weak growth (unwanted organism)

Turbidity Observed: good turbidity (wanted organism) or zero/very light turbidity (unwanted organism)

Specificity: Reject if fails to produce typical colonial morphology, size or biochemical response
Reject if fails to suppress background flora

5.7 Storage of Prepared Media

Prepared media should be stored in such a way to minimize moisture loss and provide protection against physical and microbial contamination, as well as light-induced and thermal damage. It should be stored in unopened or sealed containers at 2-8°C.

5.8 Shelf Life of Prepared Media

All prepared media should be labelled and marked with the date produced and expiry. Holding time of prepared media in different containers is listed below:

- Poured agar plates with loose-fitting covers in sealed plastic bags at 2-8°C – 2 weeks
- Large volume of agar in tightly closed screw-cap bottle at 2-8°C – 3 months
- Agar or broth in tightly closed screw-cap tubes or bottles at 2-8°C – 3 months

- a) File the list of contributors to the MU that are established to be under control and the calculations in LF W04-02 (Estimation of Measurement Uncertainty).

6. Forms

LF W09-02-01, Maintenance of Cultures

LF W09-02-02, Lists of Received Media, Reagents and Consumables

LF W09-02-03, Quality Control of Culture Media

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LF W09-02-04, Quality of Reagent Water Monitoring
LF W09-02-05, Media Volume Check
LF W04-MIC-01, Reagent and Media Preparation Logbook
LF W04-MIC-02, Analyst Worksheet for the Bacteriological Examination of Water
LF W04-MIC-03, Analyst Worksheet for the Bacteriological Examination of Fish and Fishery Products
LF W04-MIC-04, Sterility Check

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